THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 11	Put $n = 5$: $2 \times 5 + 1 = 11$.
2	**(b)** 3	Subtract any term from the next term.
3	**(a)** 22	Add the common difference 5 to the last term.
4	**(c)** 16	Multiply the last term by the common ratio 2.
5	**(c)** 2	Divide a term by the previous term.
6	**(b)** 21	$\frac{n(n+1)}{2} = \frac{6\times7}{2}$.
7	**(b)** an AP	Constant difference 2.
8	**(a)** 36	The sequence is n^2; $6^2 = 36$.
9	**(a)** 14	Put $n = 4$: $3 \times 4 + 2 = 14$.
10	**(a)** 5	Subtract any term from the next term.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$t_1 = 4$, $t_2 = 7$ (1); $t_3 = 10$, $t_4 = 13$ (1).	1+1
12	$d = 4$ (1); Next terms 22 and 26 (1).	1+1
13	$r = 5$ (1); Fourth term $= 2 \times 5^3 = 250$ (1).	1+1
14	$t_1 = 1$, $t_2 = 3$ (1); $t_3 = 5$, $t_4 = 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$t_n = a + (n-1)d$ (1); $t_{10} = 3 + 9 \times 4$ (1); $= 39$ (1).	1+1+1
16	$S_n = \frac{n(n+1)}{2}$ (1); $S = 20 \times 21 \div 2$ (1); $= 210$ (1).	1+1+1
17	$t_n = ar^{n-1}$ (1); $t_6 = 1 \times 2^5$ (1); $= 32$ (1).	1+1+1
	OR: $t_n = a + (n-1)d$ (1); $t_{10} = 5 + 9 \times 2$ (1); $= 23$ (1).	1+1+1
18	$S_n = \frac{n(n+1)}{2}$ (1); $S = 30 \times 31 \div 2$ (1); $= 465$ (1).	1+1+1
	OR: $t_n = ar^{n-1}$ (1); $t_6 = 2 \times 2^5$ (1); $= 64$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $4, 7, 10, 13, 16$ (1); Constant difference, so an AP (1); $t_{10} = 4 + 9 \times 3$ (1); $= 31$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 10/2 \times [4 + 27]$ (1); $= 5 \times 31$ (1); $S_{10} = 155$ (2).	1+1+1+2
20	Terms $7, 12, 17, 22, 27$ (1); Constant difference, so an AP (1); $t_{10} = 7 + 9 \times 5$ (1); $= 52$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 8/2 \times [10 + 28]$ (1); $= 4 \times 38$ (1); $S_8 = 152$ (2).	1+1+1+2

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 25	Add the common difference 4 to the last term.
2	**(a)** 162	Multiply the last term by the common ratio 3.
3	**(c)** 3	Divide a term by the previous term.
4	**(d)** 36	$\frac{n(n+1)}{2} = \frac{8\times9}{2}$.
5	**(b)** a GP	Constant ratio 2.
6	**(b)** 49	The sequence is n^2; $7^2 = 49$.
7	**(d)** 23	Put $n = 6$: $4 \times 6 - 1 = 23$.
8	**(d)** -4	Subtract any term from the next term.
9	**(d)** 6	Add the common difference -6 to the last term.
10	**(c)** 48	Multiply the last term by the common ratio 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d = -3$ (1); Next terms 3 and 0 (1).	1+1
12	$r = 3$ (1); Fourth term $= 4 \times 3^3 = 108$ (1).	1+1
13	$t_1 = 7$, $t_2 = 12$ (1); $t_3 = 17$, $t_4 = 22$ (1).	1+1
14	$d = 9$ (1); Next terms 44 and 53 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$t_n = a + (n-1)d$ (1); $t_{10} = 12 + 9 \times 3$ (1); $= 39$ (1).	1+1+1
16	$S_n = \frac{n(n+1)}{2}$ (1); $S = 40 \times 41 \div 2$ (1); $= 820$ (1).	1+1+1
17	$t_n = ar^{n-1}$ (1); $t_6 = 3 \times 3^5$ (1); $= 729$ (1).	1+1+1
	OR: $t_n = a + (n-1)d$ (1); $t_{10} = 2 + 9 \times 9$ (1); $= 83$ (1).	1+1+1
18	$S_n = \frac{n(n+1)}{2}$ (1); $S = 60 \times 61 \div 2$ (1); $= 1830$ (1).	1+1+1
	OR: $t_n = ar^{n-1}$ (1); $t_6 = 1 \times 4^5$ (1); $= 1024$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $2, 10, 18, 26, 34$ (1); Constant difference, so an AP (1); $t_{10} = 2 + 9 \times 8$ (1); $= 74$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 12/2 \times [2 + 22]$ (1); $= 6 \times 24$ (1); $S_{12} = 144$ (2).	1+1+1+2
20	Terms $11, 15, 19, 23, 27$ (1); Constant difference, so an AP (1); $t_{10} = 11 + 9 \times 4$ (1); $= 47$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 6/2 \times [6 + 25]$ (1); $= 3 \times 31$ (1); $S_6 = 93$ (2).	1+1+1+2

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4	Divide a term by the previous term.
2	**(b)** 78	$\frac{n(n+1)}{2} = \frac{12\times13}{2}$.
3	**(a)** neither an AP nor a GP	Differences 3, 5, 7 and ratios 4, 2.25 are not constant.
4	**(a)** 81	The sequence is n^2; $9^2 = 81$.
5	**(a)** 18	Put $n = 3$: $5 \times 3 + 3 = 18$.
6	**(c)** 7	Subtract any term from the next term.
7	**(c)** 45	Add the common difference 11 to the last term.
8	**(a)** 625	Multiply the last term by the common ratio 5.
9	**(d)** 6	Divide a term by the previous term.
10	**(d)** 120	$\frac{n(n+1)}{2} = \frac{15\times16}{2}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$r = 2$ (1); Fourth term $= 7 \times 2^3 = 56$ (1).	1+1
12	$t_1 = 7$, $t_2 = 11$ (1); $t_3 = 15$, $t_4 = 19$ (1).	1+1
13	$d = -7$ (1); Next terms 22 and 15 (1).	1+1
14	$r = 7$ (1); Fourth term $= 1 \times 7^3 = 343$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$t_n = a + (n-1)d$ (1); $t_{10} = 20 + 9 \times -2$ (1); $= 2$ (1).	1+1+1
16	$S_n = \frac{n(n+1)}{2}$ (1); $S = 75 \times 76 \div 2$ (1); $= 2850$ (1).	1+1+1
17	$t_n = ar^{n-1}$ (1); $t_6 = 5 \times 2^5$ (1); $= 160$ (1).	1+1+1
	OR: $t_n = a + (n-1)d$ (1); $t_{10} = 1 + 9 \times 6$ (1); $= 55$ (1).	1+1+1
18	$S_n = \frac{n(n+1)}{2}$ (1); $S = 200 \times 201 \div 2$ (1); $= 20100$ (1).	1+1+1
	OR: $t_n = ar^{n-1}$ (1); $t_6 = 2 \times 5^5$ (1); $= 6250$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $6, 12, 18, 24, 30$ (1); Constant difference, so an AP (1); $t_{10} = 6 + 9 \times 6$ (1); $= 60$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 14/2 \times [20 + 26]$ (1); $= 7 \times 46$ (1); $S_{14} = 322$ (2).	1+1+1+2
20	Terms $9, 11, 13, 15, 17$ (1); Constant difference, so an AP (1); $t_{10} = 9 + 9 \times 2$ (1); $= 27$ (2).	1+1+1+2
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S = 20/2 \times [8 + 114]$ (1); $= 10 \times 122$ (1); $S_{20} = 1220$ (2).	1+1+1+2

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